SPLITTING OFF THE TRUNCATIONS OF A BOUNDED ABOVE COMPLEX WHEN EXT AGAINST THE MIDDLE DEGREE VANISHES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- If Ext^{-a-1}-type connecting conditions hold, the complex splits as tau_{\leq a} K plus tau_{\geq a+1} K with the latter of finite projective amplitude. invoked at proof lines 19971-19975. Provides the initial splitting.

Source reference. Original lemma 0G97: sources/stacks-more-algebra/pinned-original-source.tex, source lines 19918–19918. Cited at proof lines 19971-19972 for the initial splitting; the conclusion is used inside the counted span.

Lemma 0.2. *Let R be a ring. Let $K \in D^-(R)$. Let $a \in \mathbf{Z}$. Assume $\mathrm{Ext}^{-a}_R(K, M) = 0$ for any R-module M. Then there is a unique direct sum decomposition $K \cong \tau_{\leq a-1}K \oplus \tau_{\geq a+1}K$ and $\tau_{\geq a+1}K$ has projective-amplitude in $[a+1, b]$ for some b.*

Proof. By Lemma 0G97 we have a direct sum decomposition $K \cong \tau_{\leq a}K \oplus \tau_{\geq a+1}K$ and $\tau_{\geq a+1}K$ has projective-amplitude in $[a+1, b]$ for some b. Clearly, we must have $H^a(K) = 0$ and we conclude that $\tau_{\leq a}K = \tau_{\leq a-1}K$ in $D(R)$. □

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